

SV-M/D: Support Vector Machine-Singular Value Decomposition Based Face Recognition

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Abstract. This paper presents a novel method for Face Recognition (FR) by applying Support Vector Machine (SVM) in addressing this Computer Vision (CV) problem. The SVM is a capable learning classifier capable of training polynomials, neural networks and RBFs. Singular Value Decomposition (SVD) are used for feature extraction and while SVM for classification. The proposed algorithm is tested on four databases, viz., FERET, FRGC Ver. 2.0, CMU-PIE and Indian Face Database. The singular values are filtered, sampled and classified using Gaussian RBF kernel for SVM. The results are compared with other known methods to establish the advantage of SV-M/D. The recall rate for the proposed system is about 90%. The colossal augmentation is due to the simple but efficient feature extraction and the ability to learn in a high dimensional space.

1 Introduction

Humans have the perpetual ability to recognize fellow beings through their faces. In Computer Vision (CV), the challenge is to make a machine (non-human) recognize a face and identify the person. Research in this field dates back to the early 70's. Through the course of time, Face Recognition has gained predominance by addressing this issue through many perspectives like Pattern Analysis, Machine Learning and Statistics. Recognition of face from static and dynamic sequences is an active area of research with numerous applications. A survey of such applications and methods are given in [5].

As shown in fig.1; two important processes that are key for a good recognition system, viz., feature extraction and classification. On the other hand, a good classifier should be able to differentiate between various face image with high accu-

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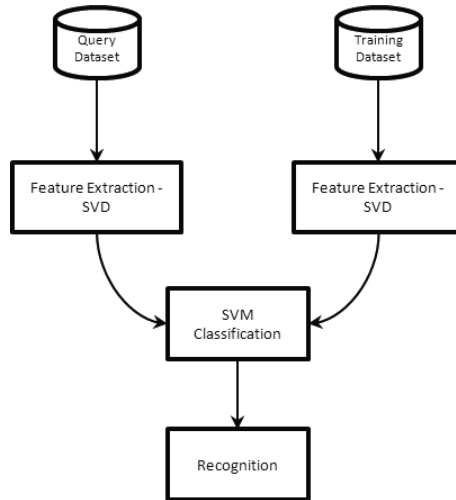


Fig. 1 An outline of the proposed system

racy and computational simplicity. In this paper we present a method using Support Vector machines (SVM) [1, 2, 3, 4] and Singular Value Decomposition (SVD) to efficiently recognize human face images. SVM is an avant garde method to train different kernels like polynomials, neural networks and Radial Basis Function (RBF) classifiers. There are several FR systems and methods developed since the inception pattern analysis. Many systems relied on the geometry like Geometrical Feature Matching. Kanade [6] in the early 70's showed that this method used the concept of matching the geometrical features based on the extraction and achieved close to 70% accuracy. Another developed method known as Eigen faces method [7, 8] used Principal Component Analysis (PCA) to project faces. These methods reported approximately 90% recognition rate on the ORL database. In summary, Eigen faces method is a fast, simple and practical method. With this came the learning perspective to FR systems. The Neural Networks (NN) have been used multiple times in tackling pattern recognition problems. This can be attributed to their non-linear mapping. The methods described in [9, 10] use Probabilistic Decision - Based Neural Network (PDBNN) for recognizing faces with a recall rate of 96%.

The recent methods are the Support Vector Machines (SVM) and Hidden Markov Model (HMM) applied to FR. The SVM is new method for training polynomial as described in [11, 12]. References [13, 14, 15, 16, 17] showed that a probabilistic Markov method would yield much better result and perform the computation in lesser amount of time. The current and contemporary researches are in Bayesian models. A comparison of SVM and HMM is presented in [18]. In [19], a new technique using a probabilistic measure of similarity, based primarily on a Bayesian (MAP) analysis was presented which resulted in a significant increase in computational speed for implementation of very large data sets.